

SECURE EMPLOYEE ACCESS AND MANAGEMENT PORTAL

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I. INTRODUCTION:

Employee management systems are essential for organizations to maintain employee records, authentication, and secure access to organizational resources. Traditional manual systems are time-consuming and less secure, which creates the need for modern web-based employee portal systems. The Secure Employee Access and Management Portal System is designed to provide a secure and centralized platform where employees can create accounts, log in securely, and access dashboard functionalities based on their role. The system ensures that only authenticated users can access protected pages.

The project also focuses on implementing cybersecurity- authentication mechanisms such as password hashing, session handling, and protected routing. The system improves user management efficiency while maintaining security and accessibility.

II. OBJECTIVE:

- To implement secure login and signup authentication.
- To provide validation for user inputs during authentication.
- To implement role-based access control for employees and administrators.
- To provide protected route access only for authenticated users.
- To develop a responsive and user-friendly employee dashboard.

III. Technologies Used

1. **React JS**
Used for developing the frontend user interface and component-based architecture.
2. **JavaScript**
Used for implementing application logic, authentication flow, and data handling.
3. **React Router DOM**
Used for page navigation and protected routing between components.
4. **Local Storage**
Used as client-side storage for storing user data, sessions, and activity logs.

5. **HTML5**

Used for creating the structure of web pages.

6. **CSS3**

Used for styling the application interface and improving user experience.

7. **Lucide React Icons**

Used for adding modern icons and improving the visual appearance of the application.

8. **SHA-256 Encryption**

Used for securely hashing passwords before storing them in local Storage.

IV. PROJECT ARCHITECTURE:

The Employee Portal System architecture is mainly focused on secure authentication and controlled user access. When a user signs up, the system validates the entered details and securely stores the password using hashing before saving it in local Storage. During login, the entered password is verified with the stored encrypted password for authentication. After successful login, a session is created to maintain user access until logout. The system also implements role-based authorization, where employees and administrators are redirected to their respective dashboards. Protected routes are used to prevent unauthorized users from accessing secured pages. Additionally, the password reset functionality allows users to securely update their password after verifying their registered email address.

Step 1 User Registration

The employee creates a new account using the signup page by entering personal details such as name, email, department, employee ID, and password.

Step 2 Password Encryption

Before storing the password, the system converts it into a SHA-256 hashed value for security purposes.

Step 3 Data Storage

The employee information and encrypted password are stored in browser local Storage using predefined storage keys.

Step 4 Login Authentication

During login, the entered password is hashed again and compared with the stored hashed password for authentication.

Step 5 Session Creation

After successful login, a session is created and stored to maintain authenticated user access.

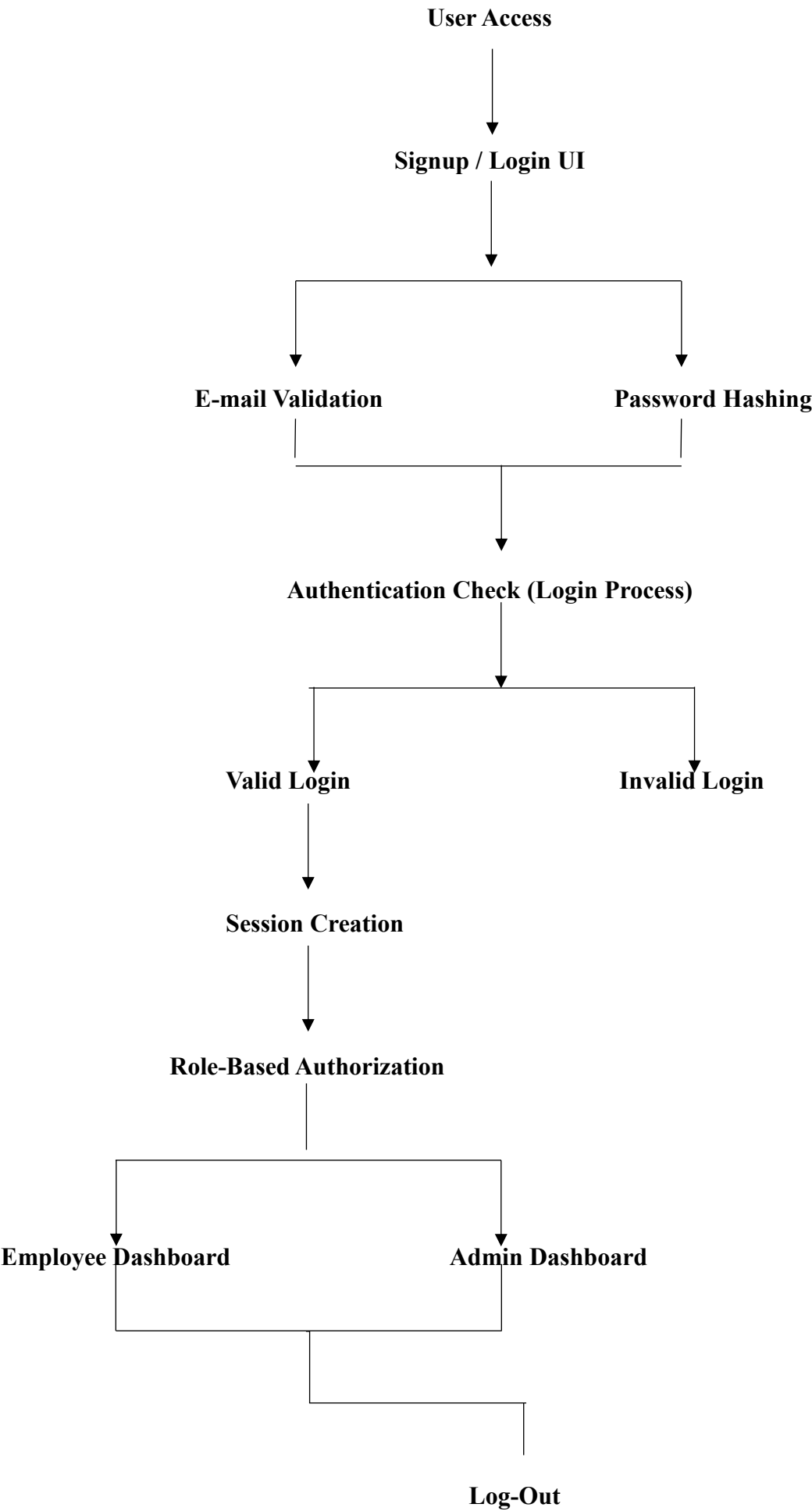
Step 6 Dashboard Access

Authenticated users are redirected to their dashboard based on their role.

Step 7 Password Reset

Employees can reset their password using the forgot password option. The new password is hashed and updated securely in local Storage.

V. SYSTEM WORKFLOW:



VI. SYSTEM PAGES AND KEY FEATURES:

SIGN IN PAGE

The Sign In page allows registered employees and administrators to securely access the portal using their login credentials. It verifies the entered details and provides protected access based on user roles.

Features:

- Email and password based secure login
- Password visibility toggle option
- Form validation for email and password fields
- Error message for invalid login credentials
- Role-based access routing (Admin / Employee)
- Session management using local Storage
- “Forgot Password” functionality for password reset

SIGN UP PAGE

The Sign-Up page is used for creating new employee accounts in the portal. It securely stores user details and validates all required information before registration.

Features:

- Employee Name input field
- Employee ID input field
- Work Email input field
- Department selection option
- Role selection option
- Strong password creation with password strength indicator
- Confirm password validation
- Duplicate email prevention
- Secure password storage using hashing
- Automatic login after successful registration

RESET PASSWORD PAGE / POPUP

The Reset Password functionality helps users recover access to their accounts by securely updating their password after verifying their registered email.

Features:

- Registered email verification
- New password creation field
- Password update functionality
- Validation for empty fields
- Error message for unregistered email
- Password stored securely in local Storage
- Success message after password reset.

ADMIN'S DASHBOARD:

The Admin Dashboard is designed for managing employees, monitoring activities, and controlling the overall workflow of the employee portal. It provides secure administrative access with role-based controls.

- Secure admin login access
- View all registered employees
- Manage employee details and roles
- Approve or reject leave requests
- Post company-wide announcements
- Monitor employee activity logs
- Access protected admin routes
- View employee department information
- Session-based authentication handling
- Secure logout functionality
- Can Inactive and Activate User

EMPLOYESS'S DASHBOARD:

The Employee Dashboard is designed to help employees securely access their personal workspace and interact with organizational services. It provides a simple and user-friendly interface for daily activities.

- Secure employee login access
- View personal profile details
- View employee ID and department
- Access role-based protected pages
- Apply for leave requests
- View leave request status
- View public announcements from admin
- Reset password functionality
- Session-based authentication support
- Secure logout functionality

VII. EMAIL AND PASSWORD VALIDATION:

The Employee Portal includes input validation to improve security and prevent invalid data entry during login and signup.

Email Validation

- Checks whether the email follows a valid email format
- Prevents empty or incorrect email entries
- Ensures users enter properly formatted work email addresses

Password Validation

- Ensures the password field is not empty
- Checks password strength for better security
- Validates minimum password requirements such as:
 - Uppercase letter
 - Lowercase letter
 - Number
 - Special character
 - Minimum length

VIII. LOCALSTORAGE USAGE

The Employee Portal uses browser local Storage to temporarily store user data, session details, and authentication information on the client side. It helps maintain user login sessions and enables quick access without using an external database.

Stored Data

- Registered user details
- Current logged-in user information
- Session and authentication status
- Activity logs and dashboard data

Advantages

- Fast data access
- Simple client-side storage solution
- Maintains login session after page refresh
- Easy integration with React authentication system

IX. SESSION MANAGEMENT

Session management is implemented to maintain secure user authentication throughout the application. After successful login, user session details are stored in local Storage and validated during navigation.

Features

- Stores active user session details
- Maintains login state after page refresh
- Automatically checks session validity
- Restricts unauthorized page access
- Clears session data during logout
- Redirects users to login after session expiration

This improves both security and user experience within the Portal system.

CONCLUSION:

The **SECURE EMPLOYEE ACCESS AND MANAGEMENT PORTAL** was successfully developed with a strong focus on authentication security and user access management. The system provides secure login, signup, password reset, role-based access control, session management, and protected navigation for both Admin and Employee users. The project also delivers a responsive and user-friendly interface that enables efficient employee management, announcement handling, and leave management while ensuring secure handling of user data and system access.